A3.2 Classification and cladistics

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-a3-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 125–137 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Classification

A3.2.1 Why classify? pp. 125–126

Millions of species need organizing. A single binomial name works in every language, groups make identification easier, and a species' group lets us predict its features.

A3.2.2 Problems with ranked taxa pp. 126–127

The ranks from kingdom to species are fixed levels, but evolution produces continuous branching. Ranks are not equivalent between groups, convergent evolution can mislead, and some traditional groups, such as reptiles, leave out descendants of their ancestor.

A3.2.3 Classifying by ancestry p. 128

If every member of a group comes from one common ancestor, members share inherited features, so we can predict the features of species that have not been studied, and the classification shows evolutionary history.

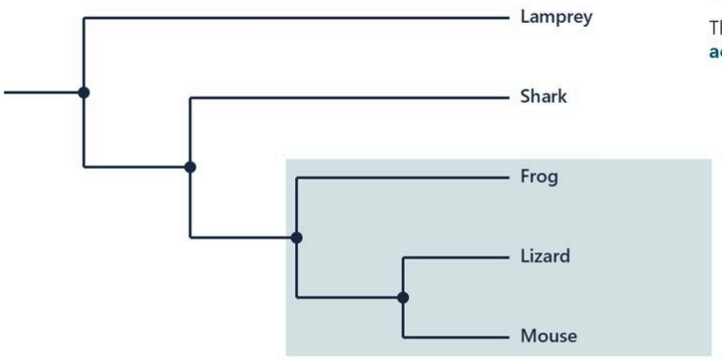
Cladistics

A3.2.4 Clades

p. 129

Clades on a cladogram

A3.2.4



Members share inherited features.

The best evidence is **base sequences** and **amino acid sequences**; morphology is used with care.

Shaded: one clade, a common ancestor and all its descendants.

Try it

Is "Shark, Frog and Lizard" a clade?

A3.2 Classification and cladistics (HL) – Textbook p. 129 12

A **clade** is a common ancestor and all its descendants. The most objective evidence for a clade comes from base sequences of genes and amino acid sequences of proteins; morphology can also be used.

A3.2.5 The molecular clock

p. 130

Sequence differences build up at a roughly steady rate, so the number of differences estimates how long ago two clades diverged. It is only an estimate because mutation rates vary with generation time, population size and selection.

A3.2.6 Building cladograms

pp. 131–133

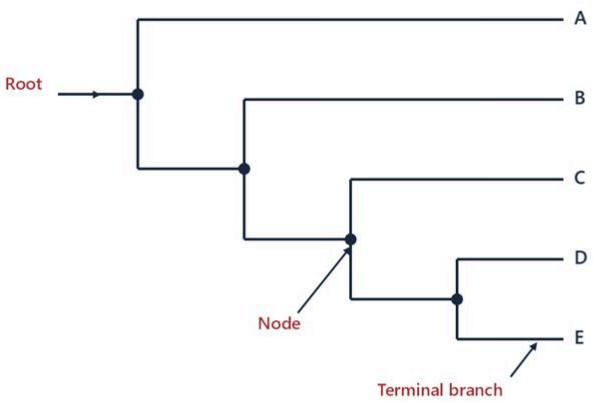
Count the differences between each pair of sequences. Join the most similar pair first, then add the others in order of similarity.

A3.2.7 Analysing cladograms

pp. 133–134

Reading a cladogram

A3.2.7



Root: shared ancestor of all.

Node: hypothetical common ancestor.

Terminal branch: one taxon.

A more recent shared node = closer relatives.

Try it

Smallest clade containing B and D? How many clades include C?

A3.2 Classification and cladistics (HL) – Textbook pp. 133–134 16

The **root** is the common ancestor of all the taxa; each **node** is a hypothetical common ancestor; each **terminal branch** leads to one taxon. Taxa that share a more recent node are more closely related.

A3.2.8 Testing classifications

pp. 135–136

DNA comparisons showed that the figwort family was not a clade, so many of its genera, such as foxgloves and speedwells, were moved to other families, mostly the plantain family.

A3.2.9 Three domains

p. 137

In 1977, ribosomal RNA sequences showed that prokaryotes form two very different groups. A new rank, the domain, now divides life into Bacteria, Archaea and Eukarya.

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